

TriNetra-AMRF: An Adaptive Multi-Modal Reliability Fusion Framework for Intelligent RGB-Thermal Surveillance and Threat Assessment

Dr. Y. D. Bhise¹

Dept. of Computer Science and Design¹

K.K Wagh Institute of Engineering Education and Research

Nashik, India

ydbhise@kkwagh.edu.in

Aditya Ahirrao², **Disha Kapse**³, **Sadique Khatib**⁴, **Vedant Sonawane**⁵

Dept. of Computer Science & Design^{2,3,4,5}

K.K Wagh Institute of Engineering Education and Research

Nashik, India

{adityaahirrao17, dishakapse55, khatibsadik9, vedantsonawane2244}@gmail.com

Abstract

TriNetra-AMRF (Adaptive Multi-Modal Reliability Fusion) is a novel deep learning framework designed for intelligent surveillance by adaptively integrating synchronized RGB and thermal imagery. Unlike conventional RGB-thermal fusion methods that combine modalities using early fusion, late fusion, attention-based, or Transformer-based mechanisms [1]–[5], the proposed framework dynamically estimates the reliability of each sensing modality according to environmental conditions, image quality, and scene characteristics before feature fusion.

The framework introduces an TriNetra Reliability Coefficient (TRC) Module that generates dynamic modality weights, enabling the fusion process to continuously adjust to variations such as illumination changes, fog, smoke, rain, glare, and thermal noise. The fused representation is processed by a multi-scale object detection network, followed by object tracking, temporal behaviour analysis, and a hierarchical threat reasoning engine capable of evaluating object behaviour over time instead of relying solely on single-frame detections.

A key contribution of the framework is the integration of perception, adaptive fusion, temporal reasoning, and explainable threat assessment into a unified pipeline. The system produces interpretable surveillance outputs consisting of detected objects, behavioural context, adaptive confidence values, threat scores, and human-readable explanations. The framework is designed for deployment in defence, aerospace, industrial monitoring, critical infrastructure protection, border surveillance, and autonomous security systems where reliable operation under diverse environmental conditions is essential [6], [7].

Keywords - RGB-Thermal Fusion, Adaptive Multi-Modal Reliability Fusion, Dynamic Reliability Weighting, Cross-Modal Feature Fusion, Computer Vision, Deep Learning, Explainable Artificial Intelligence (XAI), Real-Time Surveillance Systems.

1. Introduction

1.1 Background

Modern surveillance systems increasingly rely on multiple sensing modalities to maintain reliable operation under diverse environmental conditions. RGB cameras provide rich color, texture, and structural information during favorable lighting conditions, whereas thermal cameras capture infrared radiation independent of visible illumination, making them effective during nighttime, low-light environments, smoke, haze, and adverse weather [2], [6]. Individually, each sensing modality exhibits inherent limitations that reduce surveillance reliability under changing operational scenarios [1], [2].

Traditional RGB–Thermal fusion techniques primarily employ early fusion, late fusion, feature-level fusion, attention mechanisms, or Transformer-based architectures on the basis of some assumptions [1]–[5]. Such assumptions often result in degraded performance when one sensing modality becomes unreliable due to environmental changes or sensor degradation [3], [5].

To overcome these limitations, this work proposes TriNetra-AMRF (Adaptive Multi-Modal Reliability Fusion), a unified surveillance framework that dynamically estimates the reliability of RGB and thermal modalities before feature fusion. The framework further integrates object detection, multi-object tracking, temporal behaviour analysis, hierarchical threat reasoning, and explainable decision generation into a single intelligent surveillance architecture.

1.2 Objectives

The primary objectives of the proposed framework are:

- Develop an adaptive RGB- Thermal fusion mechanism.
- Dynamically estimate modality reliability before fusion.
- Improve robustness under varying environmental conditions.
- Integrate temporal behavioural understanding into surveillance.
- Generate explainable threat assessments instead of simple detections.
- Provide a modular framework suitable for real-time deployment.

2. Related Work

Existing RGB-Thermal fusion approaches can generally be categorized into [1] - [5]:

- Early Fusion
- Late Fusion
- Feature-Level Fusion
- Attention-Based Fusion
- Transformer-Based Cross-Modal Fusion

Although these approaches improve detection accuracy, several limitations remain [1]-[7], [10]:

- Static fusion weights assume equal modality importance.
- Environmental variations are rarely incorporated into the fusion process.
- Most systems terminate at object detection without behavioural reasoning.
- Confidence estimation is generally independent of sensing reliability.
- Explainable surveillance decisions are largely absent.

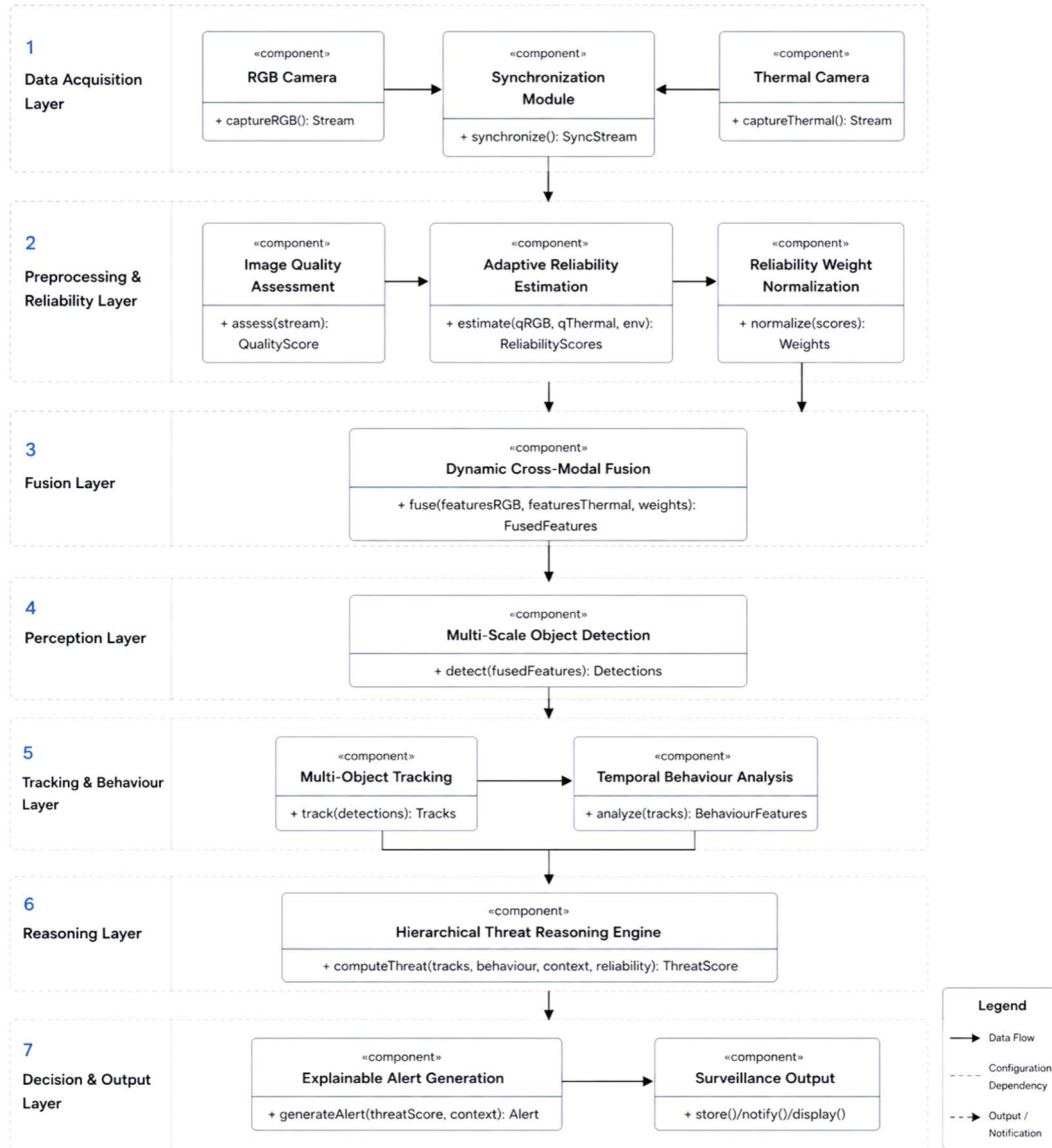
The proposed framework differs by introducing adaptive modality reliability estimation and integrating perception with temporal reasoning and explainable threat assessment.

3. Novelty Contributions

The primary contributions of TriNetra-AMRF are:

- TriNetra Reliability Coefficient (TRC) Module for dynamic modality trust assessment.
- Reliability-Guided Cross-Modal Feature Fusion.
- Integrated Multi-Object Tracking with Temporal Behaviour Analysis.
- Hierarchical Threat Reasoning Engine for intelligent surveillance.
- Explainable Alert Generation for transparent decision support.
- Detector-agnostic modular architecture supporting multiple deployment scenarios.

4. System Architecture



The proposed architecture extends conventional RGB–Thermal perception systems by integrating TriNetra Reliability Coefficient (TRC), temporal behaviour analysis, and hierarchical threat reasoning into a unified modular framework [1], [2], [6], [8].

4.1 Major Components of the Proposed Framework

1. Data Acquisition and Synchronization Module

This module acquires synchronized RGB and thermal image streams from their respective sensors. It ensures temporal and spatial alignment between both modalities before subsequent processing. Proper synchronization enables consistent feature extraction and accurate cross-modal correspondence during fusion.

2. Image Quality Assessment Module

This module evaluates the quality of each input modality independently before feature fusion. Various quality indicators are extracted to determine the reliability of RGB and thermal images under changing environmental conditions.

3. TriNetra Reliability Coefficient (TRC) Module

This module represents the primary contribution of the proposed framework. Based on image quality and environmental characteristics, it computes dynamic reliability scores that quantify the trustworthiness of RGB and thermal information.

Instead of assigning fixed importance to each modality, the framework continuously adjusts modality weights according to current sensing conditions.

4. Dynamic Cross-Modal Fusion Module

Using the adaptive reliability weights, this module combines RGB and thermal feature representations into a unified feature space. The fusion process dynamically emphasizes the modality that is more reliable while suppressing less informative features [3], [4].

5. Multi-Scale Object Detection Module

The fused feature representation is processed using a multi-scale object detection network capable of identifying surveillance objects of varying sizes and distances.

The Framework may recognize:

- Human
- Vehicle
- Aircraft
- UAV
- Equipment
- Infrastructure
- Thermal anomalies

Output

- Object classes
- Bounding boxes
- Detection confidence

7. Hierarchical Threat Reasoning Module

This module combines perception outputs, behavioural information, contextual information, and adaptive reliability scores to estimate the overall threat level associated with each tracked object.

Threat reasoning considers:

- Detected object type
- Behavioural characteristics
- Tracking consistency
- Contextual information
- Adaptive confidence

8. Explainable Alert Generation Module

The final module converts surveillance decisions into interpretable alerts suitable for human operators. Each alert includes supporting evidence that explains why a particular threat level was assigned.

5. Methodology

The proposed methodology consists of the following stages.

Stage 1: Data Acquisition

Input consists of synchronized RGB and thermal image streams.

Stage 2: Quality Assessment

Image quality metrics are extracted independently for both sensing modalities [9].

RGB metrics include:

- Brightness
- Contrast
- Sharpness

Thermal metrics include:

- Thermal variance
- Temperature consistency
- Sensor noise

Stage 3: Reliability Estimation

The framework estimates modality reliability.

$$R_{RGB} = f(Q_{RGB}, E)$$
$$R_{Thermal} = f(Q_{Thermal}, E)$$

Where

- R denotes TRC
- Q denotes image quality
- E represents environmental conditions

Normalized adaptive weights are computed as

$$W_{RGB} = \frac{e^{R_{RGB}}}{e^{R_{RGB}} + e^{R_{Thermal}}}$$
$$W_{Thermal} = 1 - W_{RGB}$$

Stage 4: Adaptive Feature Fusion

The fused feature representation is generated as:

$$F = W_{RGB}F_{RGB} + W_{Thermal}F_{Thermal}$$

This allows the framework to continuously adapt its reliance on each sensing modality [3] [4] [5].

Stage 5: Object Detection

The fused representation is processed by a multi-scale detector to identify surveillance objects [4], [8].

Stage 6: Multi-Object Tracking

Detected objects are assigned persistent identities and tracked over time [11].

Stage 7: Behaviour Analysis

Behavioural attributes include [11]:

- Velocity
- Acceleration
- Direction
- Loitering duration
- Trajectory deviation

Stage 8: Threat Assessment

Threat score is computed as:

$$Threat = f(Object, Behaviour, Context, Reliability)$$

The score is normalized between 0 and 100.

Threat Levels:

- 0–20 : Normal
- 21–50 : Observe
- 51–80 : Potential Threat
- 81–100 : Critical Threat

Stage 9: Explainable Decision Generation

The final system output contains:

- Detected object
- Object identity
- Adaptive confidence
- Behaviour summary
- Threat score
- Recommended alert level

6. Implementation Flexibility

The proposed framework is independent of any specific neural network backbone or object detector. Depending on deployment requirements and computational resources, the architecture may be implemented using end-to-end trainable dual-stream networks, transfer learning with frozen feature extractors and a trainable fusion module, or equivalent detector-specific implementations. These implementation strategies do not alter the underlying architecture or inventive concepts of the proposed framework.

7. Results (Expected Functional Outcomes)

As the proposed work describes a novel software framework intended for copyright protection, this section presents the expected functional outcomes rather than experimentally validated performance metrics.

Criterion	Conventional RGB–Thermal Fusion Frameworks	TriNetra-AMRF (Proposed Framework)
Primary Objective	Image fusion and object detection	End-to-end adaptive intelligent surveillance
Fusion Strategy	Early, Late, Feature Level, Attention or Transformer based	Adaptive Reliability-Guided Feature Fusion
Modality Weighting	Fixed or learned during training	Dynamic during inference
Image Quality Awareness	Generally Absent	Explicit quality assessment
Environmental Adaptation	Limited	Reliability-aware adaptive fusion
Multi-Object Tracking	External	Integrated
Temporal Behaviour Analysis	Rarely Included	Integrated
Threat Assessment	External or Absent	Hierarchical reasoning engine
Explainability	Limited	Explainable alerts

Framework Scope	Perception Centric	Perception + Decision Intelligence
Deployment Flexibility	Detector - dependent	Detector-agnostic modular framework

The framework is expected to:

- Dynamically adjust fusion weights according to sensing reliability.
- Improve robustness under varying illumination and environmental conditions.
- Produce more reliable detections than fixed-fusion approaches.
- Maintain object identities over long surveillance sequences.
- Detect suspicious behavioural patterns through temporal analysis.
- Generate interpretable threat scores rather than isolated detections.
- Reduce false alarms by incorporating TriNetra Reliability Coefficient (TRC).
- Support real-time deployment on intelligent surveillance platforms.

8. Discussion

The principal innovation of TriNetra-AMRF lies in integrating TriNetra Reliability Coefficient (TRC) with dynamic RGB-Thermal feature fusion, temporal behaviour modelling, and hierarchical threat reasoning into a unified surveillance framework.

Unlike existing systems that primarily emphasize feature fusion, the proposed framework extends perception into intelligent decision-making by incorporating behavioural understanding and explainable alert generation. The modular design also enables integration with different object detectors, tracking algorithms, and deployment environments without modifying the overall architecture.

Potential applications include defence surveillance, airport security, industrial inspection, border monitoring, smart city surveillance, autonomous security systems, and critical infrastructure protection.

9. Conclusion

TriNetra-AMRF presents a comprehensive adaptive multi-modal surveillance framework that combines RGB and thermal sensing with dynamic reliability estimation, feature fusion, temporal behaviour analysis, and explainable threat assessment.

By continuously adapting to sensing conditions and analysing object behaviour over time, the framework addresses several limitations of conventional RGB-Thermal fusion systems. The proposed architecture provides a scalable foundation for intelligent surveillance applications requiring robust perception, adaptive decision-making, and interpretable outputs.

10. References

Include foundational and recent literature on RGB-Thermal fusion and intelligent surveillance, for example:

- [1] J. Ma, Y. Ma, and C. Li, "Infrared and Visible Image Fusion Methods and Applications: A Survey," *Information Fusion*, vol. 45, pp. 153–178, 2019.
- [2] J. Ma, W. Yu, P. Liang, C. Li, and J. Jiang, "FusionGAN: A Generative Adversarial Network for Infrared and Visible Image Fusion," *Information Fusion*, vol. 48, pp. 11–26, 2019.
- [3] H. Li, X. Wu, and J. Kittler, "RFN-Nest: An End-to-End Residual Fusion Network for Infrared and Visible Images," *Information Fusion*, vol. 73, pp. 72–86, 2021.
- [4] Z. Zhao et al., "CDDFuse: Correlation-Driven Dual-Branch Feature Decomposition for Multi-Modality Image Fusion," *Proc. CVPR*, 2023.
- [5] W. Tang, F. He, and Y. Liu, "DATFuse: Infrared and Visible Image Fusion via Dual Attention Transformer," *IEEE Trans. Circuits Syst. Video Technol.*, 2023.
- [6] M. Brenner, N. H. Reyes, T. Susnjak, and A. L. C. Barczak, "RGB-D and Thermal Sensor Fusion: A Systematic Literature Review," *Sensors*, 2023.
- [7] Y. Luo et al., "Infrared and Visible Image Fusion: Methods, Datasets, Applications and Prospects," *Applied Sciences*, 2023.
- [8] Y. Zhang, P. Ye, H. Leung, K. Gong, and G. Xiao, "Object Fusion Tracking Based on Visible and Infrared Images: A Comprehensive Review," *Information Fusion*, 2020.
- [9] T. Alldieck, C. H. Bahnsen, and T. B. Moeslund, "Context-Aware Fusion of RGB and Thermal Imagery for Traffic Monitoring," *Sensors*, 2016.
- [10] X. Wang et al., "BDPartNet: Feature Decoupling and Reconstruction Fusion for Infrared–Visible Images," *Computers, Materials & Continua*, 2024.
- [11] M. Teutsch and W. Krüger, "Detection, Segmentation and Tracking Using Thermal Infrared Imagery: A Survey," *Machine Vision and Applications*, 2013.